

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Industry Problem Solving Task

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO4 – Precision	Assessment:	Assessment Tool 2 – Industry Problem Solving Task
Weightage:	35% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	Slot D/ 2024	Date:	27/8/26

Code of Conduct

I A Syed Thoufiq / 192472185 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: A Syed Thoufiq

Instructions

- Read each industry scenario carefully before answering.
- Analyze the given problem from a semantic analysis perspective.
- Provide structured and logical answers with proper justification.
- Support your analysis using examples from the given scenario wherever applicable.
- Use suitable diagrams, semantic representations, logical predicates, workflows, architectures, or tables whenever necessary.
- Clearly state assumptions if additional information is required.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic interpretation of user queries, identify ambiguity in meaning representation, evaluate semantic processing limitations, and improve understanding in banking chatbot applications.	Q1
Syntax-Driven Semantic Analysis	Analyze multiple interpretations of spoken commands, evaluate parsing-related ambiguity, and assess semantic extraction in real-time voice assistant systems.	Q2
First-Order Predicate Calculus, Representing Linguistically Relevant Concepts, Syntax-Driven Semantic Analysis	Design a semantic processing architecture for healthcare reports, model relationships among entities and actions, represent linguistic concepts, and improve semantic extraction accuracy.	Q3

Annexure A – Industry Problem Solving Task

1. AI-Powered Insurance Claim Processing System

An insurance company is developing an NLP-based system to automatically process customer insurance claims. The system uses **Context-Free Grammar (CFG)** to parse customer statements and extract information such as the claimant, damaged object, cause of damage, and claim amount.

However, the system faces the following problems:

- Ambiguous sentence structures
- Difficulty distinguishing noun and prepositional phrase attachment
- Subject–verb agreement errors
- Inefficient parsing of lengthy customer descriptions
- Difficulty handling conversational and incomplete input

Example customer statement:

"I reported the damage to the car after the accident."

The sentence can produce different interpretations depending on whether "**to the car**" is associated with *reported* or *damage*.

Tasks:

- a) Analyze the structural ambiguity in the given sentence using CFG and illustrate the possible interpretations.
- b) Evaluate the limitations of a basic CFG in handling ambiguity, agreement, and long customer-generated insurance statements.
- c) Design an improved parsing approach using **PCFG, Feature Structures, and Earley Parsing** to resolve ambiguity and efficiently process the claim.
- d) Justify how the proposed architecture can improve the accuracy, scalability, and reliability of automated insurance claim processing.

2. Intelligent Railway Ticket Booking Assistant

A railway company is developing a conversational AI assistant for ticket booking. The system currently uses **top-down parsing** to interpret passenger requests.

The system encounters:

- Excessive backtracking
- Ambiguous attachment of phrases
- Incomplete passenger requests
- Long-distance dependencies
- Slow response for complex queries

Example passenger request:

"Book two tickets from Chennai to Delhi for my parents with AC sleeper."

The system sometimes incorrectly associates "**with AC sleeper**" with the passengers instead of the ticket/class information.

Tasks:

- a) Analyze the possible syntactic interpretations of the given passenger request using CFG.
- b) Evaluate why top-down parsing may result in excessive backtracking and inefficient processing for conversational ticket-booking queries.
- c) Analyze how **Earley Parsing** can handle ambiguity, incomplete input, and different possible parse structures more effectively.
- d) Compare **Top-Down Parsing, CFG-based parsing, and Earley Parsing** in terms of ambiguity handling, computational efficiency, partial-input processing, and suitability for real-time railway booking systems.

3. AI-Based Legal Contract Analysis System

A legal technology company is developing an NLP system to analyze contracts and automatically identify:

- Parties involved
- Obligations
- Conditions
- Deadlines
- Actions
- Exceptions

The existing system uses a basic CFG parser. However, it produces incorrect interpretations when processing long legal sentences containing relative clauses, passive constructions, and multiple prepositional phrases.

Example contract sentence:

"The supplier who delivered the equipment to the company must replace the defective components within thirty days."

The system sometimes incorrectly identifies the **company** as the entity responsible for delivering the equipment.

Tasks:

- a) Analyze the syntactic structure of the given sentence using CFG and identify possible sources of ambiguity.
- b) Evaluate why basic CFG is insufficient for accurately interpreting complex legal sentences containing relative clauses and nested structures.

c) Design an NLP architecture integrating **CFG, PCFG, Feature Structures, and Earley Parsing** to improve syntactic and semantic interpretation.

d) Develop a step-by-step processing workflow showing how the system can extract:

- Supplier
- Action
- Equipment
- Company
- Obligation
- Deadline

and justify how the proposed approach improves automated contract analysis.

4. AI-Based E-Commerce Product Search System

An e-commerce company is developing a conversational product-search system. Customers enter natural-language queries rather than selecting predefined filters.

The current system uses CFG but faces difficulties with:

- Ambiguous phrase attachment
- Coordination
- Missing words
- Informal customer queries
- Long product descriptions
- Real-time search requirements

Example user query:

"Find laptops with a touchscreen for students under ₹60,000."

The system may incorrectly associate "**for students**" with the touchscreen rather than the laptop.

Tasks:

a) Analyze the syntactic ambiguity in the given query and construct possible parse interpretations using CFG.

b) Evaluate the limitations of CFG in handling informal, elliptical, and ambiguous e-commerce search queries.

c) Propose an improved parsing architecture using **PCFG, Feature Structures, and Earley Parsing** to identify product attributes, user requirements, and price constraints.

d) Justify how the proposed solution can improve search precision, response time, and customer experience in a large-scale e-commerce platform.

5. Intelligent Airline Voice Assistant

An airline is developing a real-time voice assistant that allows passengers to modify flight bookings using spoken commands.

The system currently uses a conventional top-down parser. It experiences:

1. Backtracking
2. Ambiguous attachment
3. Speech recognition errors
4. Incomplete commands
5. Delayed responses
6. Difficulty interpreting corrections made by users

Example spoken command:

"Change my flight to Mumbai tomorrow with two passengers."

Depending on the parse, "with two passengers" may be incorrectly interpreted as an attribute of the destination rather than the number of passengers.

The passenger may also say:

"Change my flight to Mumbai tomorrow... actually, make that Bangalore."

Tasks:

- a) Analyze the possible syntactic structures and ambiguities in the given voice command.
- b) Evaluate the limitations of top-down parsing when processing incomplete, corrected, and ambiguous spoken commands in real time.
- c) Design a parsing architecture using Earley Parsing, PCFG, and Feature Structures to support partial input, ambiguity resolution, and agreement constraints.
- d) Develop a processing workflow showing how speech input can be converted into a structured booking request and justify the proposed method for real-time airline applications.

Rubrics for Calculation

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding ; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Q. No. / Criteria Level	Understanding of Industry Problem	Application of Engineering Concepts	Solution Design	Use of Tools	Analysis	Sub. Total	Total %
1	3	3	4	4	4	18	92
2	3	4	4	4	4	19	
3	4	4	3	4	4	19	
4	3	4	3	4	4	18	
5	3	4	4	3	4	19	

Faculty In-charge	Course Coordinator	Program Director
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Date: 27/8/26	Date: _____	Date: _____

Question 1 – AI-Powered Insurance Claim Processing System

The insurance system uses CFG to parse customer statements and extract the claimant, damaged object, cause of damage and claim amount. The supplied example is: “I reported the damage to the car after the accident.”

CFG Representation

S → NP VP

NP → PRP

VP → V NP PP

NP → Det N

PP → P NP

PP → P NP

A simplified grammar can generate a structure in which the prepositional phrase “to the car” is attached inside the noun phrase headed by “damage” or attached to the reporting verb phrase. This creates structural ambiguity.

Possible Interpretations

Interpretation	Structure	Meaning
A	VP → V [NP the damage] [PP to the car] [PP after the accident]	The reported event concerns damage, and 'to the car' identifies the affected/damaged object.
B	VP → V [NP the damage [PP to the car]] [PP after the accident]	'to the car' is attached to the noun 'damage', specifying damage to the car.

For an insurance system, Interpretation B is especially useful because it links the damaged object directly to the damage concept. However, a basic CFG can generate both structures unless additional probabilistic or semantic information is used.

Limitations of Basic CFG

1. A basic CFG can generate multiple valid parses without ranking them.
2. It does not inherently resolve noun-phrase versus verb-phrase prepositional attachment.
3. Agreement constraints such as subject–verb number are not represented by simple nonterminals.
4. Long customer descriptions can create many possible parse trees and increase parsing cost.
5. Conversational and incomplete input may not match strict grammar productions.
6. Semantic information such as claimant, damaged object and cause is not automatically encoded by CFG.

Improved Architecture: PCFG + Feature Structures + Earley

Customer Statement

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Text Normalization

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v

CFG Grammar + Feature Structures

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v

Earley Parsing

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+--> Multiple Parse Candidates

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v

PCFG Probability Ranking

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v

Semantic Role / Entity Extraction

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v

Claim Record

(Claimant, Object, Cause, Amount)

|

v

Validation + Claim Processing

Earley Parsing is suitable for the scenario because it can handle general CFGs and retain partial states while parsing. PCFG probabilities can rank competing parses, while feature structures can encode agreement and grammatical features.

Feature Structure Example

NP:

CATEGORY = NP

NUMBER = singular

SEMANTICS = damaged_object(car)

VP:

CATEGORY = VP

SUBJECT = claimant

EVENT = report

OBJECT = damage

TARGET = car

Expected Benefits

Requirement	Basic CFG	Improved Approach
Ambiguity	Generates alternatives	PCFG ranks alternatives
Agreement	Weakly represented	Feature structures encode constraints
Long input	Can be inefficient	Earley supports chart-based parsing
Incomplete input	Poor fit	Earley chart can retain partial

		states
Semantic extraction	Separate processing needed	Semantic features/roles can be integrated
Reliability	More false interpretations	Ranking + constraints + validation reduce errors

The architecture can improve accuracy by selecting a higher-probability interpretation, scalability by avoiding repeated parsing work through chart parsing, and reliability by validating grammatical and semantic constraints before the claim is processed.

Question 2 – Intelligent Railway Ticket Booking Assistant

The railway assistant uses top-down parsing for passenger requests. The supplied example is: “Book two tickets from Chennai to Delhi for my parents with AC sleeper.”

CFG Interpretation

S → VP

VP → V NP PP PP PP

NP → Num N

PP → P NP

NP → Det N

NP → ProperNoun

NP → PossDet N

NP → PrepClass

The phrase “with AC sleeper” is ambiguous because it can be associated with the passenger phrase “my parents” or with the ticket/travel class. In the intended railway-booking interpretation, “AC sleeper” describes the ticket/class requested.

Interpretation	Attachment of “with AC sleeper”	Result
A – Intended	Ticket / travel-class information	Two AC-sleeper tickets for the parents
B – Alternative	Passenger phrase “my parents”	AC sleeper is incorrectly treated as a property of the passengers

Why Top-Down Parsing Backtracks

1. Top-down parsing begins from the start symbol and predicts productions before seeing all input evidence.
2. When a chosen production later fails, the parser may return to an earlier decision.
3. Ambiguous attachments create multiple possible production choices.
4. Long requests contain more points where an early choice can become invalid.
5. Conversational input may be incomplete, forcing the parser to explore alternatives before enough information is available.

Earley Parsing for the Scenario

Earley Parsing maintains a chart of partial states and can represent incomplete constituents while scanning the sentence. It is therefore better suited to ambiguous and partially observed conversational input than a naive top-down strategy that repeatedly restarts alternatives.

For a production $S \rightarrow \alpha \cdot \beta$:

- dot before β = symbols still expected
- completed state = complete constituent
- chart[i] stores states at input position i

This allows:

1. Prediction of possible rules
2. Scanning of matching words
3. Completion of recognized constituents
4. Preservation of competing parses

Comparison

Factor	Top-Down Parsing	CFG-Based Chart Approach	Earley Parsing
Ambiguity	May cause repeated backtracking	Can represent CFG alternatives	Maintains competing states
Backtracking	Potentially high	Can be reduced with charting	Avoids repeated work through chart states
Incomplete input	Weak	Depends on implementation	Can preserve partial states
Complex queries	May become slow	Better with chart methods	Designed for general CFG parsing
Real-time booking	Less suitable for ambiguity-heavy input	Moderate	More suitable when optimized

Recommendation

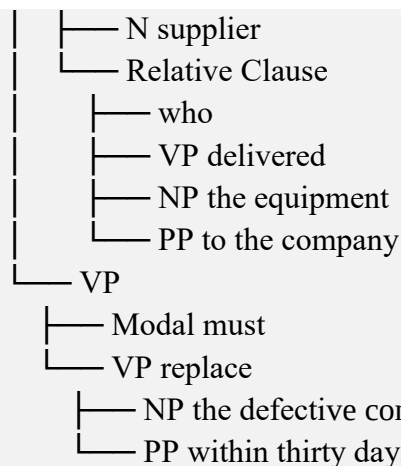
For a railway booking assistant, Earley Parsing combined with feature constraints and probabilistic ranking is preferable to a simple top-down parser. The system can then interpret 'with AC sleeper' as a class constraint rather than a passenger attribute.

Question 3 – AI-Based Legal Contract Analysis System

The legal NLP system must identify parties, obligations, conditions, deadlines, actions and exceptions. The example is: “The supplier who delivered the equipment to the company must replace the defective components within thirty days.”

CFG Structural Analysis

S
├── NP
│
└── Det The



The relative clause “who delivered the equipment to the company” modifies “the supplier.” The phrase “to the company” is attached to “delivered” and indicates the destination/recipient context of the delivery. The main obligation is “must replace the defective components,” with the deadline “within thirty days.”

Main Ambiguity and Error

The reported error is that the system identifies the company as the entity responsible for delivering the equipment. This is incorrect. The relative pronoun “who” refers back to “the supplier,” so the supplier is the delivery actor.

Entity / Phrase	Correct Role
supplier	Party / Agent of delivery; party with replacement obligation
who	Refers to supplier
equipment	Object / Theme of delivery
company	Destination/Recipient of delivery
defective components	Object of replacement
thirty days	Deadline

Why Basic CFG Is Insufficient

1. CFG represents syntactic structures but does not inherently encode semantic reference such as who the relative pronoun refers to.
2. Nested clauses create several possible constituent structures.
3. Multiple prepositional phrases can produce attachment ambiguity.
4. Passive constructions and long-distance dependencies require additional feature or semantic information.
5. Legal language often contains long sentences with multiple obligations and conditions.

Integrated Architecture

Legal Contract

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Sentence Segmentation

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CFG Candidate Generation

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v

Feature Structure Constraints

(number, agreement, clause features)

|
v

Earley Parsing

|
v

PCFG Parse Ranking

|
v

Coreference + Semantic Role Analysis

|
v

Legal Event Extraction

|
v

Structured Contract Record

Step-by-Step Extraction

Step	Operation	Output
1	Tokenize and segment	Supplier / delivered / equipment / company / replace / thirty days
2	Parse main clause	Supplier must replace components
3	Identify relative clause	who delivered equipment to company
4	Resolve 'who'	who → supplier
5	Extract delivery event	supplier = agent; equipment = object; company = destination
6	Extract obligation	supplier must replace defective components
7	Extract deadline	within thirty days
8	Build structured record	Party + action + object + deadline

Structured Output

SUPPLIER = supplier

DELIVERY_AGENT = supplier

DELIVERY_OBJECT = equipment

DELIVERY_DESTINATION = company

OBLIGATION:

PARTY = supplier

ACTION = replace

OBJECT = defective components

MODALITY = must

DEADLINE = thirty days

This design reduces incorrect responsibility assignment by combining syntactic structure with feature constraints and semantic interpretation. PCFG ranking helps select among competing parses, while Earley Parsing provides a general chart-based mechanism for the complex grammar.

Question 4 – AI-Based E-Commerce Product Search System

The e-commerce system processes natural-language product queries. The example is: “Find laptops with a touchscreen for students under 60,000.”

Possible CFG Interpretations

S → VP

VP → V NP

NP → NBAR PP PP PP

NBAR → N

PP → with NP

PP → for NP

PP → under PRICE

NP → Det N

NP → Adj N

Interpretation	Attachment	Meaning
A – Intended	“for students” modifies laptops	Find laptops intended/suitable for students that have a touchscreen and cost under ₹60,000
B – Alternative	“for students” modifies touchscreen	Find laptops with a touchscreen intended for students, where the phrase is treated as a property of the touchscreen

The first interpretation is the natural product-search representation because the user requirement “for students” describes the target laptop rather than the touchscreen itself.

Limitations of CFG

1. Informal queries may omit grammatical words.
2. Elliptical queries may contain fragments rather than complete sentences.
3. Coordination can create multiple structural interpretations.
4. CFG alone does not assign probabilities to competing interpretations.
5. Long product descriptions can increase the number of candidate parses.
6. A fixed grammar may not cover new product names, abbreviations and colloquial expressions.

Improved Parsing Architecture

Natural-Language Query

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v

Normalization / Query Segmentation

|

v

CFG Grammar + Feature Structures

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v

Earley Parsing

|

v

PCFG Ranking

|

v

Semantic Slot Extraction

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+--> Product = laptop

+--> Attribute = touchscreen

+--> User Requirement = students

+--> Price Constraint = < ₹60,000

|

v

Search / Ranking Engine

Feature Structure

QUERY:

PRODUCT = laptop

ATTRIBUTE:

touchscreen = required

USER:

students

PRICE:

operator = <

value = 60000

currency = INR

Benefits

Goal	Improvement
Search precision	Semantic slots distinguish product attributes from user requirements
Ambiguity	PCFG ranks likely attachment structures
Partial/informal input	Earley can preserve partial chart states

Scalability	Chart parsing avoids repeatedly recomputing the same substructures
Customer experience	More accurate results reduce irrelevant products

The final search representation should treat laptop as the product, touchscreen as a product attribute, students as a target-user requirement and ₹60,000 as a price constraint.

Question 5 – Intelligent Airline Voice Assistant

The airline assistant receives spoken commands to modify flight bookings. The supplied command is “Change my flight to Mumbai tomorrow with two passengers.” The passenger may also correct the destination mid-utterance: “Change my flight to Mumbai tomorrow... actually, make that Bangalore.”

Possible Syntactic Structures

Interpretation	Attachment of “with two passengers”	Meaning
A – Intended	Booking/request	Change the flight to Mumbai tomorrow for two passengers
B – Incorrect	Destination phrase	Treat 'two passengers' as a property attached to Mumbai

The phrase should be represented as a passenger-count constraint rather than as an attribute of the destination.

Correction Handling

In the corrected command, the later phrase “actually, make that Bangalore” supersedes the earlier destination Mumbai. A robust system therefore needs dialogue-state updating rather than simply concatenating the two destinations.

Initial interpretation:

DESTINATION = Mumbai

Correction detected:

"actually, make that Bangalore"

Updated dialogue state:

DESTINATION = Bangalore

PASSENGER_COUNT = 2

DATE = tomorrow

ACTION = change flight

Limitations of Top-Down Parsing for Voice Input

1. Speech recognition errors can create tokens that do not match grammar rules.
2. Incomplete utterances can leave a top-down parser without a complete derivation.
3. Corrections can invalidate earlier parse decisions.
4. Ambiguous attachments require exploration of multiple alternatives.
5. Repeated backtracking can increase response latency.

6. Spoken commands often contain disfluencies, repetitions and self-corrections.

Proposed Architecture

Speech Signal

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Automatic Speech Recognition

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Text + Confidence Scores

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v

Earley Parser

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+--> Partial / Alternative Parse States

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v

PCFG Ranking

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v

Feature Structure Validation

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v

Dialogue-State / Correction Handler

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v

Structured Booking Request

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v

Booking Modification API

Feature Structure

BOOKING_CHANGE:

ACTION = change

DESTINATION = Bangalore

DATE = tomorrow

PASSENGER_COUNT = 2

CONSTRAINTS:

PASSENGER_COUNT = integer

DATE = temporal_expression

DESTINATION = location

Step-by-Step Workflow

Step	Processing	Output
1	Receive speech	Audio command
2	ASR transcription	Change my flight to Mumbai tomorrow with two passengers
3	Earley parse	Candidate syntactic structures
4	PCFG ranking	Most probable interpretation
5	Feature validation	Destination/location and passenger-count constraints
6	Correction detection	Mumbai replaced by Bangalore
7	Dialogue-state update	Destination = Bangalore
8	Structured request	Change flight, Bangalore, tomorrow, 2 passengers
9	Execution / confirmation	Booking modification request

Real-Time Benefits

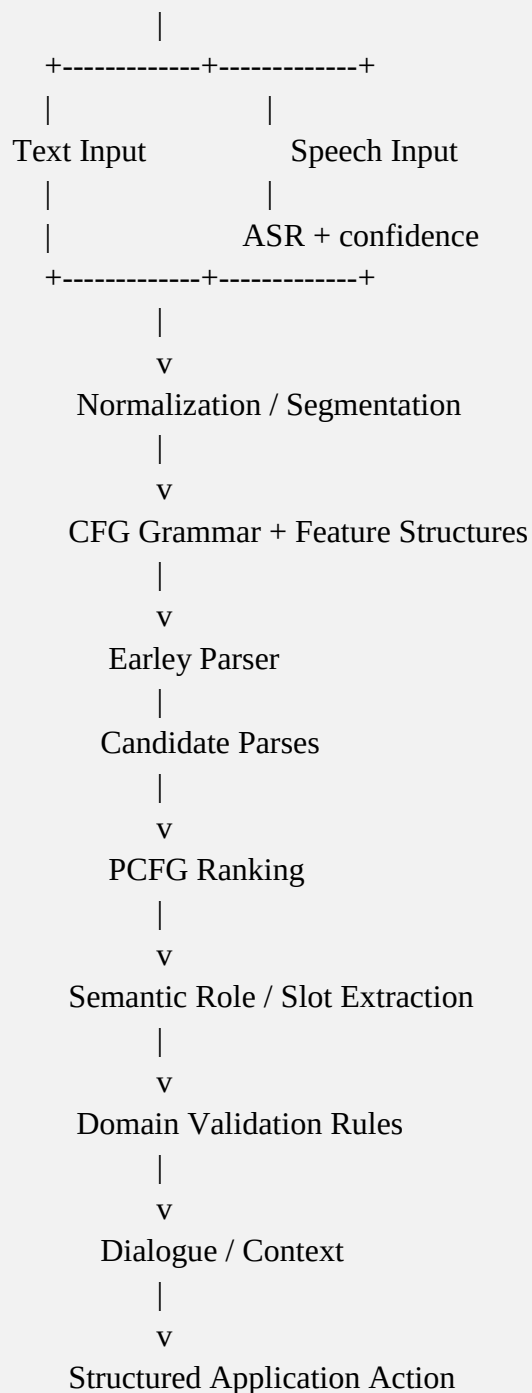
1. Earley parsing supports ambiguous CFG parsing without committing immediately to one structure.
2. PCFG ranking can prioritize likely interpretations.
3. Feature structures enforce type and agreement constraints.
4. Dialogue-state handling allows corrections to overwrite earlier values.
5. ASR confidence can be used to request confirmation when recognition is uncertain.
6. The final structured representation separates language understanding from booking execution.

Comparative Analysis of Proposed Solutions

Application	Main Ambiguity	Recommended Core Approach	Key Benefit
Insurance	PP attachment	CFG + PCFG + Feature Structures + Earley	Better claim-role extraction
Railway	Class phrase attachment	Earley + probabilistic ranking	Reduced backtracking
Legal	Relative clauses / nested PPs	CFG + PCFG + Feature Structures + Earley	Correct responsibility and obligation extraction
E-Commerce	Attribute/user-requirement attachment	CFG + PCFG + Feature Structures + Earley	Higher search precision
Airline Voice	Attachment + correction + incomplete input	Earley + PCFG + Feature Structures + dialogue state	Real-time robust command interpretation

Overall Architecture Recommendation

User Input



This combined architecture is appropriate across the five scenarios because CFG supplies explicit syntactic rules, feature structures constrain grammatical properties, Earley Parsing manages general CFG ambiguity and partial structures, and PCFG provides probabilistic ranking. Domain-specific semantic validation and dialogue state then convert the selected interpretation into an application-level action.

Conclusion

The five industry scenarios demonstrate that a basic CFG is useful for defining syntactic structures but is not sufficient by itself for modern conversational, legal, insurance, commerce and voice applications. Ambiguous attachment, incomplete input, long-distance relationships, speech errors and complex nested structures require additional parsing and representation mechanisms.

PCFG adds probabilistic ranking, Feature Structures provide grammatical constraints and attribute-value information, and Earley Parsing provides a general chart-based approach for ambiguous and partial input. Together, these techniques provide a practical foundation for reliable parsing frameworks aligned with the assessment's CO5 statement.